

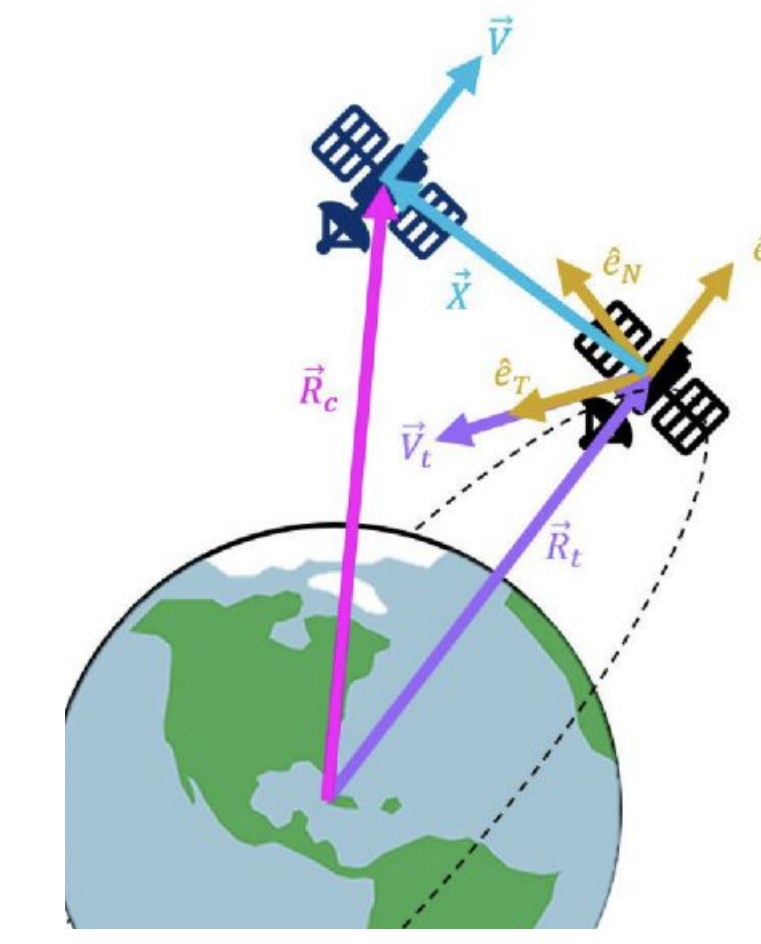


REACH YOUR GOALS AND AVOID UNCERTAINTY: CERTIFIED NEURAL LYAPUNOV-BARRIER CONTROL FOR UNKNOWN DYNAMICAL SYSTEMS

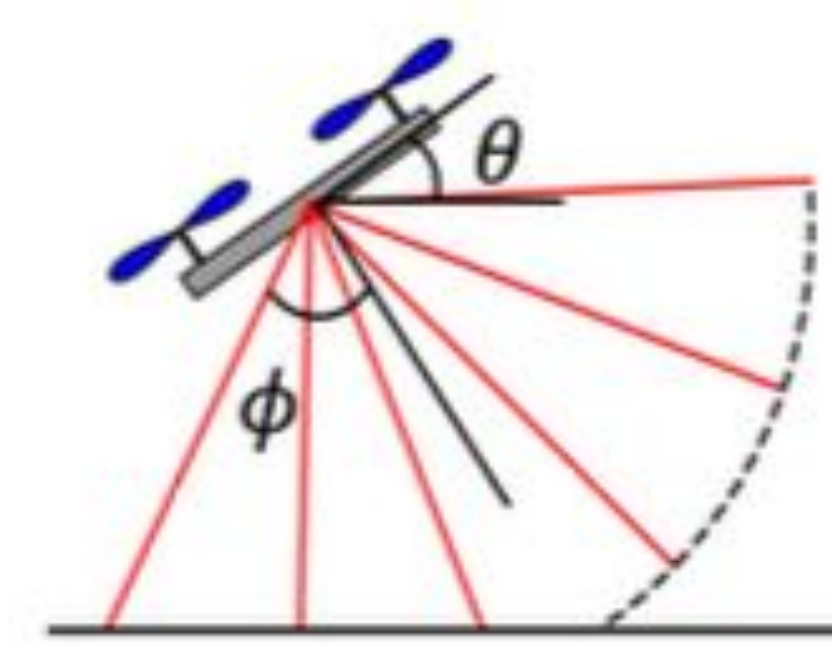
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Background and Motivation

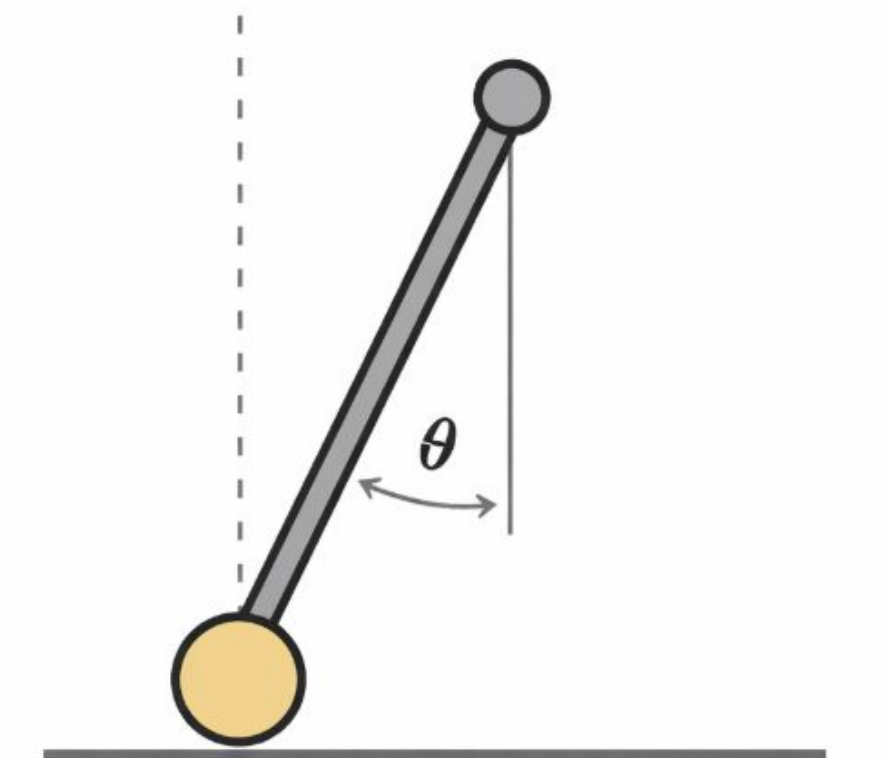
- Reinforcement Learning neural controllers achieve strong performance but **lack formal safety and liveness guarantees**
- **Motivation:** Existing certificate-based methods (Lyapunov / barrier) assume **known system dynamics**
 - Real-world systems often have **unknown / black-box dynamics**
- Our idea:** Learn a **surrogate model** of the dynamics from environment data
- Use it to:
 - synthesize controller and certificate
 - verify safety properties
- **Challenge:** Guarantees may not transfer to the real system



Inverted Pendulum



2D Docking



2D Quadrotor

Methodology

1. Learn surrogate dynamics

- Sample transitions from black-box system
- Train a neural network approximation of dynamics

2. Controller & Certificate Synthesis

- Counterexamples Guided Inducted Synthesis (CEGIS)
 - Train on samples
 - Verify and obtain counter examples
 - Repeat until no counterexamples are found

3. Verification

- Ensure initial set is under safe threshold

$$\forall x \in \mathcal{X}_I. \quad V(x) \leq \beta$$

- Prove descent condition holds in the domain

$$\forall x \in \mathcal{X} \setminus \mathcal{X}_G. \quad V(x) \leq \beta \rightarrow V(x) - V(f(x, \pi(x))) \geq \epsilon$$

4. Transfer Guarantees through Lipschitz Analysis

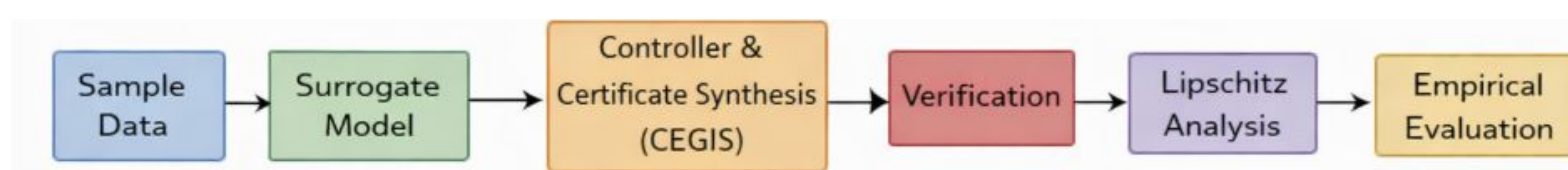
- Bound error between true and learned dynamics
- Determine sampling density required so that a robust certificate exists based on the one trained

$$\epsilon = \epsilon_r - L_\infty \cdot \delta > 0$$

Where L_∞ is Lipschitz of certificate, δ is the desired bound, and ϵ_r is the certified descent parameter

$$h \leq \frac{2(\delta - M)}{L_f + L_g},$$

where h is sampling spacing, L_f, L_g are Lipschitz constants, M is max error, and δ is the desired bound



Experimental Setup

- Perform CEGIS with surrogate dynamics for three control scenarios
- Simulate 5,000 trajectories starting in initial region
- Count number of safety and liveness violations

2D Docking

Controller	# Docks	# Unsafe	Avg. Steps	Max Steps
Initial	5,000	549	6.07 ± 1.81	11
Trained (True)	5,000	0	15.40 ± 7.25	28
Trained (Surrogate)	5,000	0	10.66 ± 3.99	23

Inverted Pendulum

Controller	# Docks	# Unsafe	Avg. Steps	Max Steps
Initial	5,000	0	9.68 ± 0.77	11
Trained (True)	5,000	0	11.25 ± 4.06	22
Trained (Surrogate)	5,000	0	57.29 ± 24.52	94

2D Quadrotor

Controller	# Docks	# Unsafe	Avg. Steps	Max Steps
Initial	5,000	2,735	786.3 ± 203.5	1,179
Trained (True)	5,000	2,291	347.8 ± 125.7	642
Trained (Surrogate)	4,990	2,265	225.0 ± 70.6	389

Findings

- Certified controllers can be learned using **data-driven surrogate models**
- Conditional guarantee assuming true dynamic Lipschitz
- Guarantees depend on:
 - Lipschitz smoothness bounds
 - Surrogate maximum error
- Amount of points required to sample in practice is **infeasible** (might work for lower dimensions)
- **BUT:** promising empirical performance safety performance on real systems